Modeling Neutron Stars
Lecture-4

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UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
January 16, 2021

Contents:

Elementary Field Theory

- Recap
- Noether's Theorem Applications
- Equation of motion

Classical Fields

E-L Equations and Hamiltonian

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- Euler-Lagrange Equations in particle mechanics: $\boxed{\frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{q}_r}\right) = \frac{\partial \mathrm{L}}{\partial q_r}}$

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$$\boxed{\mathscr{H}(\Phi^{\mathrm{A}}, \boldsymbol{\nabla}\Phi^{\mathrm{A}}, \Pi_{\mathrm{A}}) = \Pi_{\mathrm{A}}\dot{\Phi}^{\mathrm{A}} - \mathscr{L}(\Phi^{\mathrm{A}}, \partial_\mu \Phi^{\mathrm{A}}).}$$

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- We can define a *Noether's charge* which remains conserved as,

$$\boxed{Q \equiv \int d^3x J^0}$$

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Problem 1: Show P^0 is same as the total Hamiltonian defined before. Find the stress-energy tensor for the real scalar, complex scalar and the electromagnetic field with the Lagrangians given in Problem 2 of Lec-3.

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Problem

- **Problem 2: Consider the Lagrangian for complex scalar field $\mathscr{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - m^2 \phi^\dagger \phi - V(\phi^\dagger \phi)$. The Lagrangian remains invariant under $\phi \to e^{-iq\theta}\phi, \phi^\dagger \to e^{iq\theta}\phi^\dagger$.**
  (a)Use Noether's theorem to find the corresponding conserved current j^μ. Verify $\partial_\mu j^\mu = 0$.
  (b) Calculate the conserved Noether current for the transformation when the electromagnetic field is also included and the combined Lagrangian is given by

$$\begin{aligned} \mathscr{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \quad &+ \quad [(\partial^\mu - iqA^\mu)\phi^\dagger][(\partial_\mu + iqA_\mu)\phi] \\ &- \quad m^2 \phi^\dagger \phi - V(\phi^\dagger \phi). \end{aligned}$$

  (c) The Euler-Lagrange equations for A_μ should give the Maxwell's equations. Show that the current appearing in Maxwell's equations is same as in part (b).

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where, $\Box \equiv \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \boldsymbol{\nabla}^2 = \partial_\mu \partial^\mu$.

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Equation of motion for scalar field

Klein-Gordon equation (small note on quantization)

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Equation of motion for Dirac field

Constructing a Hamiltonian

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Equation of motion for Dirac field

Properties of γ matrices

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Equation of motion for Dirac field

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- Combining these together one can write,

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Equation of motion for Dirac field

Properties of γ matrices

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- Combining these together one can write,

$$[\gamma^\mu, \gamma^\nu]_+ = 2g^{\mu\nu}.$$

- **Problem 3: Show that all the γ^μ matrices are traceless.**

Equation of motion for Dirac field

Matrices from γ matrices

Equation of motion for Dirac field

Matrices from γ matrices

- Define a set of matrices $\sigma^{\mu\nu}$ from γ matrices as,

$$\sigma^{\mu\nu} = \frac{i}{2}\left[\gamma^{\mu}, \gamma^{\nu}\right]_{-} = -\sigma^{\mu\nu}.$$

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- Define γ^5 matrix as,

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Equation of motion for Dirac field

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$$\epsilon_{0123} = 1.$$

Equation of motion for Dirac field

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- **Problem 4: Show that γ^5 is traceless and it satisfies $\left[\gamma^{\mu}, \gamma^5\right]_{+} = 0$. Show that $\left(\gamma^{i}\right)^{\dagger} = -\gamma^{i}$ and $(\gamma^{\mu})^{\dagger} = \gamma^0\gamma^{\mu}\gamma^0$.**

Dirac Equation

Covariant form

Dirac Equation

Covariant form

- Using the Dirac Hamiltonian, equation of motion for a field ψ is written as,

$$i\frac{\partial}{\partial t}\psi(x) = \gamma^0(-i\boldsymbol{\gamma}\cdot\boldsymbol{\nabla} + m)\psi(x)$$

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$$(i\gamma^\mu\partial'_\mu - m)\psi'(x') = 0$$

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$$\Rightarrow i\gamma^\mu\Lambda_\mu{}^\nu\partial_\nu S(\Lambda)\psi(x) - mS(\Lambda)\psi(x) = 0$$

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$$\begin{aligned} \Rightarrow i\gamma^\mu\Lambda_\mu{}^\nu\partial_\nu S(\Lambda)\psi(x) - mS(\Lambda)\psi(x) &= 0 \\ \Rightarrow S^{-1}(\Lambda)\gamma^\mu\Lambda_\mu{}^\nu S(\Lambda) &= \gamma^\nu \end{aligned}$$

Lorentz transformation

Dirac equation

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- Infinitesimal Lorentz transformation is given by

$$\Lambda_{\mu\nu} = g_{\mu\nu} + \omega_{\mu\nu}, \text{ with } \omega_{\mu\nu} = -\omega_{\nu\mu}.$$

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$$S^{-1}(\Lambda)\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}S(\Lambda) \quad = \quad \gamma^{\rho}$$

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- Now we start from

$$\begin{aligned} S^{-1}(\Lambda)\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}S(\Lambda) &= \gamma^{\rho} \\ \left(1 + \frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right)\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}\left(1 - \frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right) &= \gamma^{\rho} \end{aligned}$$

Lorentz transformation

Dirac equation

- Continuing....

$$\left(1+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right)\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}\left(1-\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right)=\gamma^{\rho}$$

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Dirac equation

- Continuing....

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$$\Rightarrow \quad \gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}-\frac{i}{4}\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}\beta_{\mu\nu}\omega^{\mu\nu}=\gamma^{\rho}$$

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$$-\frac{i}{4}\gamma^{\lambda}g^{\rho\kappa}(g_{\lambda\kappa}+\omega_{\lambda\kappa})\beta_{\mu\nu}\omega^{\mu\nu}=\gamma^{\rho}$$

Lorentz transformation

Dirac equation

- Continuing....

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Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
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\Rightarrow\quad & \gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}-\frac{i}{4}\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}\beta_{\mu\nu}\omega^{\mu\nu}=\gamma^{\rho} \\
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\Rightarrow\quad & g^{\rho\kappa}\gamma^{\lambda}(g_{\lambda\kappa}+\omega_{\lambda\kappa})+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\gamma^{\lambda}g^{\rho\kappa}(g_{\lambda\kappa}+\omega_{\lambda\kappa}) \\
& \qquad -\frac{i}{4}\gamma^{\lambda}g^{\rho\kappa}(g_{\lambda\kappa}+\omega_{\lambda\kappa})\beta_{\mu\nu}\omega^{\mu\nu}=\gamma^{\rho} \\
\Rightarrow\quad & \gamma^{\rho}+g^{\rho\kappa}\gamma^{\lambda}\omega_{\lambda\kappa}+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}-\frac{i}{4}\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}\beta_{\mu\nu}\omega^{\mu\nu}=\gamma^{\rho} \\
\Rightarrow\quad & 4ig^{\rho\kappa}\gamma^{\lambda}\omega_{\lambda\kappa}-\left(\beta_{\mu\nu}\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}-\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}\beta_{\mu\nu}\right)\omega^{\mu\nu}=0
\end{aligned}
$$

Lorentz transformation

Dirac equation

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$$
\begin{aligned}
& \left(1+\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right)\gamma^{\lambda}\Lambda_{\lambda}{}^{\rho}\left(1-\frac{i}{4}\beta_{\mu\nu}\omega^{\mu\nu}\right)=\gamma^{\rho} \\
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\Rightarrow\quad & 4ig^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu}\omega^{\mu\nu}=\left(\beta_{\mu\nu}\gamma^{\rho}-\gamma^{\rho}\beta_{\mu\nu}\right)\omega^{\mu\nu}
\end{aligned}
$$

Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
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\Rightarrow\quad & 4ig^{\rho\kappa}\gamma^{\lambda}\omega_{\lambda\kappa}-\left(\beta_{\mu\nu}\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}-\gamma^{\lambda}g^{\rho\kappa}g_{\lambda\kappa}\beta_{\mu\nu}\right)\omega^{\mu\nu}=0 \\
\Rightarrow\quad & 4ig^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu}\omega^{\mu\nu}=\left(\beta_{\mu\nu}\gamma^{\rho}-\gamma^{\rho}\beta_{\mu\nu}\right)\omega^{\mu\nu} \\
\Rightarrow\quad & 2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu}-g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu}=[\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu}
\end{aligned}
$$

Lorentz transformation

Dirac equation

- Continuing....

$$2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu} - g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu} = [\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu}$$

Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
& 2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu} - g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu} = [\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu} \\
\Rightarrow \quad & 2i\left(\delta^{\rho}_{\nu}\gamma_{\mu} - \delta^{\rho}_{\mu}\gamma_{\nu}\right) = [\beta_{\mu\nu},\gamma^{\rho}]_{-}
\end{aligned}
$$

Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
& 2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu} - g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu} = [\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu} \\
\Rightarrow \quad & 2i\left(\delta^{\rho}_{\nu}\gamma_{\mu} - \delta^{\rho}_{\mu}\gamma_{\nu}\right) = [\beta_{\mu\nu},\gamma^{\rho}]_{-} \\
\Rightarrow \quad & 2ig^{\rho\tau}\left(g_{\tau\nu}\gamma_{\mu} - g_{\tau\mu}\gamma_{\nu}\right) = g^{\rho\tau}\,[\beta_{\mu\nu},\gamma_{\tau}]_{-}
\end{aligned}
$$

Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
& 2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu} - g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu} = [\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu} \\
\Rightarrow\quad & 2i\left(\delta^{\rho}_{\nu}\gamma_{\mu} - \delta^{\rho}_{\mu}\gamma_{\nu}\right) = [\beta_{\mu\nu},\gamma^{\rho}]_{-} \\
\Rightarrow\quad & 2ig^{\rho\tau}\left(g_{\tau\nu}\gamma_{\mu} - g_{\tau\mu}\gamma_{\nu}\right) = g^{\rho\tau}\,[\beta_{\mu\nu},\gamma_{\tau}]_{-}
\end{aligned}
$$

- So we can write in a compact form,

$$
\boxed{[\gamma_{\tau},\beta_{\mu\nu}]_{-} = 2i\left(g_{\tau\mu}\gamma_{\nu} - g_{\tau\nu}\gamma_{\mu}\right)}
$$

Lorentz transformation

Dirac equation

- Continuing....

$$
\begin{aligned}
& 2i\left(g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\mu}g_{\kappa\nu} - g^{\rho\kappa}\gamma^{\lambda}g_{\lambda\nu}g_{\kappa\mu}\right)\omega^{\mu\nu} = [\beta_{\mu\nu},\gamma^{\rho}]_{-}\,\omega^{\mu\nu} \\
\Rightarrow\quad & 2i\left(\delta^{\rho}_{\nu}\gamma_{\mu} - \delta^{\rho}_{\mu}\gamma_{\nu}\right) = [\beta_{\mu\nu},\gamma^{\rho}]_{-} \\
\Rightarrow\quad & 2ig^{\rho\tau}\left(g_{\tau\nu}\gamma_{\mu} - g_{\tau\mu}\gamma_{\nu}\right) = g^{\rho\tau}\,[\beta_{\mu\nu},\gamma_{\tau}]_{-}
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- So we can write in a compact form,

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$$

- **Problem 5: Show that above relation holds for $\beta_{\mu\nu} = \sigma_{\mu\nu}$ which is defined as $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^{\mu},\gamma^{\nu}]_{-}$.**

References:

- A first book of Quantum Field Theory, CRC Press, **Amitabha Lahiri and Palash Baran Pal**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.